

FATEMEH POURKHAVARI

Khavari Unit 293, Gosarjeva ulica 9, Ljubljana 1000, Slovenia

Phone: +386 69 706 201

Email: f.pourkhavari@gmail.com

LinkedIn Profile: [linkedin.com/in/fatemeh-pourkhavari-b16665173](https://www.linkedin.com/in/fatemeh-pourkhavari-b16665173)

RESEARCH INTERESTS

Entanglement, Cavities, Nonlinear optics, Quantum optics, Plasmonics, Graphene, Nanoparticles

EDUCATION

SEPTEMBER 2016 – DECEMBER 2018

MASTER'S DEGREE IN PHYSICS MOLECULAR ATOMIC, UNIVERSITY OF RAZI, IRAN

GPA (in original scale): 16.24 out of 20 | Converted GPA: 81.2 out of 100

Thesis Title: Magneto-Optical Sensor Based on Optical Bistability Phenomena in Nonlinear Nanoparticle.

First Supervisor: Dr. Nader Daneshfar, Second Supervisor: Dr. Tayebbeh Naseri

FEBRUARY 2012 – AUGUST 2015

BACHELOR'S DEGREE IN PHYSICS, MALAYER UNIVERSITY, IRAN

GPA (in original scale): 17.56 out of 20 | Converted GPA: 87.8 out of 100

Thesis Title: The effects of electromagnetic field

Supervisor: Dr. AliReza Abdykian

RESEARCH EXPERIENCE

2023 – PRESENT

RESEARCH ASSISTANT, LJUBLJANA UNIVERSITY, FACULTY OF MATHEMATICS AND PHYSICS, SLOVENIA

Supervisor: Dr. Rainer Kaltenbaek

1. Worked on the development of a narrowband entanglement source for quantum optics experiments

2019 – 2022

RESEARCH ASSISTANT, RAZI UNIVERSITY, IRAN

Supervisor: Dr. Tayebbeh Naseri

1. Mathematical analysis and MATLAB simulations to:
 - design optical devices such as biosensors and optical tweezers
 - understand the concept of efficient broadband absorption of infrared radiation in thin layers
 - study coherent perfect absorption in suspended monolayer graphene
 - understand quantum optics, photonics, one-dimensional photonic crystal (1DPC) and nonlinear optics
 - study quantum hydrodynamic model surface plasmon at an interface between a monolayer graphene sheet to use it in a project

2. Studying photonic methods for measuring vibration in gas turbines.

INDUSTRY EXPERIENCE

2022 – 2023

RESEARCH TRAINEE AND DEVELOPMENT, AZMON SANAT
NAZERAN TADBIR (NON-DESTRUCTIVE TESTING), IRAN

RESEARCH PROJECTS

2023-PRESENT

LJUBLJANA UNIVERSITY, FACULTY OF MATHEMATICS AND PHYSICS,
SLOVENIA

A Compact Source of Narrowband Entanglement for Cesium-Based Quantum Memories.

- Quantum Optics, Entanglement, Narrowband Source

2019

UNIVERSITY OF RAZI, IRAN

Study of optical pulling force and plasmonic resonance in core-shell spherical nanoparticles.

- Nanomaterials, Optical pulling force
- Scattering theory, Surface plasmons

2019 - 2020

UNIVERSITY OF RAZI, IRAN

Photonic Methods for Precision Measurements in Gas Turbines.

- Gas turbine, Photonic methods
- Accurate measurement, Vibration measurement

2020 - 2021

UNIVERSITY OF RAZI, IRAN

Study of Efficient Broadband Nanoparticle Absorbers in Infrared Optical Frequency Range.

- Absorption, Scattering, Metal nanoparticles
- Quasi-static approximation, Surface plasmons

2020 - 2021

UNIVERSITY OF RAZI, IRAN

Study of Coherent Perfect Absorption in Multilayer Nanostructure Coated with Graphene.

- Coherent Perfect Absorption (CPA), Graphene, Incident Angle
- TE mode, TM mode

SKILLS

- Programming skills: PYTHON , KICAD, SOLID WORK, MATLAB, MATHEMATICA, LATEX
- Computation skills: Microsoft Office package: Microsoft Word, Excel, Access, MathType
- Language skills: 1) Kurdish (mother tongue)
2) Persian (native)

- 3) English (Fluent) | IELTS exam taken on **APRIL 2022**
- 4) German (A2)
- 5) Slovenian (A1)
- Interpersonal skills: Project management | Strong decision maker | Innovative | Service - focused | Complex problem solver

SUMMER SCHOOL

BARI, SEPTEMBER 2024

QUANTUM 2024 – SUMMER SCHOOL ON QUANTUM OPTICAL TECHNOLOGIES, BARI, ITALY

attended the aforementioned Summer School

WORKSHOPS, INTERNSHIPS, HONORS AND AWARDS

JUNE 2017

FACULTY OF ENGINEERING, UNIVERSITY OF RAZI, IRAN

Certificate of participation in MATLAB education workshop

JUNE 2018

FACULTY OF SCIENCE, UNIVERSITY OF RAZI, IRAN

Certificate of participation in MATLAB education workshop

JUNE 2018

FACULTY OF SCIENCE, UNIVERSITY OF RAZI, IRAN

Certificate of participation in English language workshops and understanding of scientific physics text

SEPTEMBER 2018

FACULTY OF SCIENCE, UNIVERSITY OF RAZI, IRAN

Certificate of participation in physics specialty subject workshop

PUBLICATIONS AND CONFERENCES

2018

1. Tayebe Naseri, Nader Daneshfar and **Fatemeh Pourkhavari**, “Investigation of optical bistability in core-shell nanoparticles consisting of nonlocal core-shell magnetoplasmonic shell”, Journal of Materia physics of plasmas **25**, 082304-(2018). (<https://doi.org/10.1063/1.5024781>)
2. **Fatemeh Pourkhavari**, Tayebah Naseri, Nader Daneshfar (2018, May), “Optical bistability in core-shell magnetoplasmonic nanoparticles”, National conference on Micro/Nanoscale Technologies. (<https://civilica.com/doc/807669/>)
3. **Fatemeh Pourkhavari**, Nader Daneshfar, Tayebah Naseri (2018, August), “Study of optical scattering from core shell cylindrical and spherical nanoparticles by considering magneto-optic property”, the annual physics conference of Iran. (PACS No. **42.65.E s**)

2019

4. Tayebe Naseri and **Fatemeh Pourkhavari**, “Impact of geometry and magneto-optical properties on field enhancement and optical bistability in core-shell nanoparticles”, Journal of Theoretical and Applied Physics, 13(1), 23-29- (2019). (<https://doi.org/10.1007/s40094-018-0314-z>)

2020

5. Tayebe Naseri and **Fatemeh Pourkhavari**, “Bimetallic Core-Shell with Graphene Coating Nanoparticles Enhanced Optical Properties and Slow Light Propagation”, Journal of Material plasmonics, 1-8-(2020). (<https://doi.org/10.1007/s11468-019-01101-w>)

REFERENCES

- Dr. Rainer Kaltenbaek | rainer.kaltenbaek@fmf.uni-lj.si
- Dr. Nader Daneshfar | ndaneshfar@razi.ac.ir
- Dr. Tayebeh Naseri | tnaseri@razi.ac.ir
- Dr. AliReza Abdykian | abdykian@malayeru.ac.ir
- Dr. Masoud Rezvani Jalal | revanijalal@gmail.com